

Final Project Deliverable 1 - Project Proposal

Contrastive Representation Learning for Blood-Brain Barrier Permeability Prediction

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1. Choice of Dataset

Reasons for selection:

1. Feasibility for Contrastive Learning: For our primary "pre-training" phase (Contrastive Learning), we require a large corpus of chemical structures without needing specific bioactivity labels. We will use a representative subset of the **ZINC dataset** (approximately 100k molecules) for pretraining. ZINC is specifically curated for "drug-like" commercially available compounds, meaning the SMILES strings are cleaner and require less domain-specific filtering than raw ChEMBL data.
2. Standardized Benchmarking: For the evaluation phase (Section 2c), we will use **MoleculeNet datasets** (specifically **BBBP** for blood-brain barrier penetration). These are widely used benchmark datasets for molecular property prediction, allowing us to directly compare our model's performance against established baselines.
3. Risk Mitigation: Unlike scraping raw ChEMBL, which requires complex SQL queries and normalization of experimental units, these datasets are available in ready-to-use formats (CSV/SDF), allowing us to focus our efforts on the complex architecture (CNN/RNN + Contrastive Loss) rather than data engineering.

2. Methodology

a. Data Preprocessing

Feasibility & Approach: The dataset is feasible as it consists primarily of text-based chemical representations (SMILES) which are lightweight to store and process.

Preprocessing Steps:

1. **Standardization:** We will use **RDKit** to canonicalize all SMILES strings (e.g., removing salts, standardizing tautomers) to ensure a unique text representation for every molecule.
2. **Feature Extraction (View 1 - Graph/Vector):** We will compute **Morgan Fingerprints (ECFP4)** and RDKit physicochemical descriptors (e.g., MolWt, LogP) using **rdkit**. This serves as the "structural/vector" view of the molecule.
3. **Sequence Tokenization (View 2 - Sequence):** For the RNN/CNN input, SMILES strings will be tokenized (character-level or atom-level) and padded to a fixed length.
4. **Augmentation (If time permits):** Explore simple SMILES augmentation techniques to generate positive pairs for the contrastive objective.

b. Machine Learning Model

Goal: We aim to learn a robust, unified molecular representation (embedding) by contrasting two different "views" of the same molecule: the 1D sequence view (SMILES) and the explicit structural view (Fingerprints). This effectively allows the deep learning model to "understand" chemical structure by aligning it with pre-calculated descriptors.

Proposed Architecture: Dual-Encoder Contrastive Framework (Inspired by DrugCLIP)

Our model will feature a two-branch "Siamese-style" architecture:

1. **Branch A (Sequence Encoder): A 1D-CNN or RNN (LSTM/GRU).**
 - *Input:* Tokenized SMILES sequences.
 - *Role:* To capture the sequential syntax and local substructure patterns within the text string.
2. **Branch B (Feature Encoder): A Multi-Layer Perceptron (MLP).**
 - *Input:* Morgan Fingerprints and RDKit descriptors.
 - *Role:* To encode global structural features and presence/absence of functional groups.
3. **Contrastive Loss Head:** Both branches project their outputs into a shared latent space (e.g., 128-dim vector). We will use **InfoNCE loss (NT-Xent)** to maximize the cosine similarity between the SMILES and Fingerprint representations of the *same* molecule

(positive pairs) while minimizing similarity to *other* molecules in the batch (negative pairs).

Reasoning:

- **CNN/RNN:** Excellent for processing sequential data like SMILES text.
- **Contrastive Learning:** This allows us to leverage large amounts of unlabelled data (self-supervised learning). By forcing the sequence encoder to match the fingerprint encoder, we ground the "black box" neural network in established chemical theory (represented by the fingerprints).

Alternatives Considered:

- *Graph Neural Networks (GNNs):* While GNNs are the standard for molecules, they are computationally heavier. Our proposed CNN/RNN approach on SMILES is faster and lighter, making it easier to deploy in a web app.

c. Evaluation Metric

Since this is primarily a representation learning task followed by a downstream prediction task, we will use two sets of metrics:

1. **Pre-training Metric (Retrieval Task):** We will evaluate embedding alignment using Top-K retrieval accuracy. Given a SMILES embedding, we measure whether the matching fingerprint embedding of the same molecule is among the top K most similar representations within a batch (and vice versa), indicating successful cross-view alignment.
2. **Downstream Task (Property Prediction):** We will freeze (or fine-tune) the pre-trained encoder and add a linear classifier to predict BBB permeability using the BBBP dataset.
 - **ROC-AUC:** For binary classification (Does this molecule cross the blood-brain barrier? Yes/No).

Baseline to beat: We aim to beat a standard Random Forest model trained solely on raw fingerprints.

3. Application

Concept: A "Molecular Translator and Property Predictor" Web App.

User Input:

- **Text Input:** Users can type or paste a SMILES string.
- **Visual Input (If time permitting):** We can embed a simple JSME (JavaScript Molecule Editor) canvas allowing users to draw a molecule, which converts the drawing to a SMILES string automatically.

System Output:

1. **Latent Visualization:** A 2D PCA/t-SNE plot showing where this molecule sits in the "chemical space" relative to known drugs (e.g., "Your molecule is structurally similar to Aspirin").
2. **Property Prediction:** The model will output predicted BBB permeability properties based on the learned representation.

Integration: We will use **Streamlit** for the backend to load the PyTorch model and serve the predictions to a simple frontend.